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Network Clock Frequency Synchronization

Omer Gurewitz*, Israel Cidon and Moshe Sidi
Electrical Engineering Department
Technion, Haifa 32000
Israel

Abstract

The emergence of network convergence emphasizes the need to support distributed synchronous servers such as TDMoIP (pseudo-wire) and 3G cellular gateways over a packet switched infrastructure. Consequently, we formalize the problem of network wide clock frequency synchronization and introduce novel and efficient algorithms to synchronize the frequency among all the clocks in the network with respect to a single frequency.

The common thread of our solutions is that they take a network-wide view that accounts for all the clocks in the network and measurements taken over all links to estimate the frequency difference of each clock with respect to the reference clock. The various presented algorithms introduce different trade-offs between the accuracy and the computation complexity. While all our schemes are global, they employ simple pair-wise measurements between neighboring nodes. Consequently, all the algorithms presented in the paper, are simple, easy to implement and require a modest amount of measurement and control traffic.

I. Introduction

Clock synchronization is a basic requirement for distributed computation infrastructures. The correctness and performance level of many distributed applications depend on the quality of the clocks' synchronization between the entities that assemble the system. It is known that clocks within general purpose computers are usually low cost and inaccurate. Due to the quality of its crystal oscillator a typical computer clock can gain or lose 10 seconds per day which makes a simple wristwatch a much better timekeeper [1].

In this study we focus on the frequency synchronization aspects of the general task of Clock Synchronization. Frequency synchronization had a great impact on the evolution of traditional telecommunication, especially in enabling the transition from analog to digital transmission and switching. A survey on the importance of network synchronization over the past 30 years is given in [2]. As we will show in the sequel, frequency synchronization is still a crucial issue in topical terrestrial and cellular converged networks.

Traditional PSTN (Public Switched Telephone Network) use frequency synchronization in order to multiplex a number of signals on the same line. The required accuracy of the synchronization increases with the number of signals multiplexed on a single line [3].

The development of Synchronous Digital Hierarchy (SDH) and Synchronous Optical Network (SONET) technologies introduced new and more stringent demands on network synchronization [4]. Note that in TDM networks, network wide frequency synchronization is crucial to meet proper BER as TDM data may be inserted and dropped at each link. In particular, switching TDM data streams from an incoming link to another outgoing link at a node requires tight synchronization between the frequencies of the two links and consequently between any two non-neighboring nodes. Therefore, traditional TDM networks implement the frequency synchronization using a dedicated synchronization structure usually based on a tree topology and operates at the physical layer using dedicated signals.

Recently, there has been a new interest in TDM services implemented over IP backbones (TDMoIP, also termed pseudo-wire). TDMoIP is a transport technology that extends traditional TDM

* Currently with the Electrical and Computer Engineering Department at Rice University, Houston, TX.

circuits across packet-switched cores transparently and economically. Obviously, TDMoIP, like any other TDM, requires frequency synchronization among all TDM gateways in order to operate properly. Cellular networks and in particular third generation systems (3G) also require network wide frequency synchronization platforms. Such platforms facilitate interoperability between different systems such as 2G and 3G. They also facilitate interfacing to the PSTN/circuit switched networks, improve successful call handovers and synchronizing spatial reuse of time division multiplexing slots. Both TDMoIP and next generation 3G networks (B3G) are expected to be deployed over IP infrastructures. Hence the nodes that require to be frequency synchronized create isolated TDM islands placed within a packet based core. The problem in such a structure is that the clock frequency synchronization between the TDM entities can no longer rely on a dedicated synchronization network built upon dedicated signals, indeed it must use packet exchange to synchronize the different clocks in the overlay network.

Several works have addressed frequency synchronization in packet networks. However, these works limit the frequency synchronization to a single pair of nodes. The main issue addressed in most works is the filtering of the influence of clock frequencies differences from delay measurements [5–7]. Another set of works that deal with a pair-wise frequency synchronization is related to the playback of real-time traffic. For example, the ATM adaptation layers AAL1 and AAL2 in [8] attempt to synchronize the frequency of a sending node with the frequency of a receiving node using information embedded in the packets header. Similar to the above, these techniques are limited to a single node pair synchronization. Techniques for network-wide synchronization have been developed in the area of time offset synchronization [9–11]. However, none have tackled the aspect of frequency synchronization.

Extending the nodes pair-wise frequency synchronization to a network-wide synchronization so that all clocks in the network will run with the same frequency is not a straight forward task. Even if each node maintains all the pair-wise frequency differences with respect to all its neighbors it is difficult to understand how the node should adjust its single operational frequency using all pair-wise differences. A simple and intuitive approach for extending the pair-wise frequency synchronization into a network-wide synchronization scheme is by using an NTP like hierarchical approach. According to this approach, an overlay hierarchical subnetwork of clock servers is maintained. Each clock server estimates its frequency difference with respect to a single clock server which is closer to a reference clock server (the root of the overlay network). There are several drawbacks to the hierarchical approach. First, it is not clear that the closer a server is to the reference time server, its frequency is also closer to the reference clock frequency. Furthermore, according to a hierarchical scheme, two neighboring servers that have the same hop distance from the reference time server (not necessarily through the same parent node) will adjust their clocks regardless of the link connecting them. This can result in a considerably large frequency difference due to different measurement noise characteristics along the two different paths to the reference time server.

The main contribution of this paper is the introduction of a new and improved way (as compared to the intuitive hierarchical approach described above) to synchronize the frequency among all the clocks in the network with respect to a single reference frequency. In this work we suggest schemes for a global clocks frequency synchronization in a packet based network. The common thread of our schemes is that they offer network-wide approaches that take into account all the clocks in the network as well as all measurements taken over all links and estimate the frequency difference of each clock with respect to the reference frequency. The trade-offs between the various schemes is in their complexity and accuracy. While all the schemes are global, they employ only pair-wise measurement probes between neighboring nodes, the same probes developed for synchronizing a single pair of nodes. Consequently, all the algorithms presented in the paper, are simple, easy to

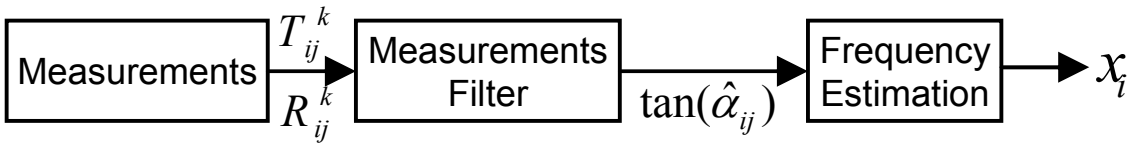


Fig. 1. The Network-Wide procedure for estimating the frequency difference between each node and the reference frequency.

implement and do not trigger an excessive amount of traffic.

The global procedures for estimating the frequency difference between each node and the reference frequency work as follows: we start by collecting a set of measurements between each neighboring pair. Next, based on the measurements, we use a measurement filter to estimate the frequency difference between each pair of neighboring clocks. Finally we use all pair-wise estimations to estimate each frequency and adjust the clocks accordingly (see Figure 1). The measurements collection is explained in Section II-B. In section III we explain the operation of the measurement filter, and in sections IV and V we explain the global procedure for estimating actual frequency of each node and its adjustment. Finally, in section VI we provide numerical results, compare the different global procedures and demonstrate their major advantages over the intuitive hierarchical approach in a practical parameter setup.

II. The Model

This study suggests a network-wide approach for synchronizing all clocks in the network to run with the same frequency. We start by introducing the model used throughout the paper. We split the model description into two aspects: the network and the measurements. We begin by introducing the network model that is used.

A. The Network Model

Our network model is an overlay network, whose nodes are the entities that need to be synchronized, and its links are the connections among them; by a connection we refer to directed paths comprising one or more physical links connecting two synchronization concerned entities without passing through any other such entity on the way.

The entities which are meant to be frequency synchronized are termed nodes and are denoted by Λ_i for node i. $\mathcal{N}$ denotes the set of all nodes and $N = |\mathcal{N}|$ is the number of nodes. By Λ_0 we denote the reference node to which we expect all the nodes in the network to be synchronized with and we term its frequency as the *reference frequency*. We define a directed link between two nodes as a directed path between the two nodes that does not contain any other node in $\mathcal{N}$. Since the physical path connecting node Λ_i to node Λ_j is not necessarily the same as the one connecting node Λ_j to node Λ_i we will distinguish between the link connecting nodes Λ_i and Λ_j which will be denoted by e_{ij}, and the one connecting nodes Λ_j and Λ_i which will be denoted by e_{ji}. The collection of all links will be denoted by $\mathcal{E}$. Although we distinguish between e_{ij} and e_{ji} we will assume throughout the paper that all links are bidirectional, namely if $e_{ij} \in \mathcal{E}$, then $e_{ji} \in \mathcal{E}$ (if e_{ij} exists so does e_{ji}). Finally, denote by G_i the set of nodes which are node Λ_i's neighbors in the overlay network, and let $|G_i|$ be the number of such neighbors.

B. The Measurements

Extensive literature exists on how to conduct network measurements. Most employed schemes are based on an active measurement systems that measure round-trip and one-way delays over various Internet paths [12–15]. While systems like NTP uses periodic trigger of measurement probes, all

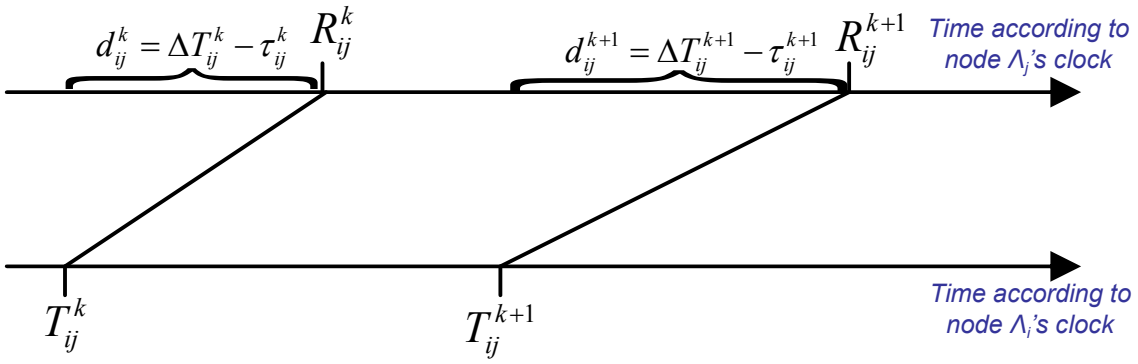


Fig. 2. Two probes sent by Λ_i to Λ_j such that $e_{ij} \in \mathcal{E}$.

active systems quoted above, schedule measurements at Poisson intervals. In order to synchronize their clocks, nodes compare their current clocks with their neighbors via probe exchanges. Since timing information tends to deteriorate while travelling through the network, delay characteristics will highly influence the accuracy of the synchronization. In this subsection we describe the probes exchanged between two neighboring nodes, and review the delay characteristics of each such probe. Throughout the explanation we concentrate on a single pair of neighboring nodes Λ_i and Λ_j, assuming the same is true for all neighboring pairs.

Node Λ_i is continuously sending probes (at Poisson or periodic times) to node $\Lambda_j \in G_i$. Time is stamped on probe k by the sender Λ_i upon transmission (T_{ij}^k). The receiver Λ_j stamps its local time upon receiving the probe (R_{ij}^k). Note that NTP in which probes are sent round trip collects all four time stamps on each probe. Therefore the NTP probe format can be used [16]. Unlike NTP, our method treats one-way probes independently from each other. The probe exchange is illustrated in Figure 2.

We divide the delay experienced by a probe into two basic components, constant and variable. The constant part relates to all parts of the link delay which are deterministic, e.g., transmission delay and propagation delay. The variable component, is usually associated with packet queueing and processing. This part may vary from probe to probe even when the probes have the same size and format. Let us denote by d_{ij}^k the delay experienced by probe k while travelling on the link e_{ij}, and by c_{ij}^k and v_{ij}^k its constant and variable parts, respectively ($d_{ij}^k = c_{ij}^k + v_{ij}^k$). Since the deterministic part is constant for all probes traveling the link we will omit the superscript for the delay constant part, i.e. $c_{ij}^k = c_{ij}\ \forall k$.

Let us also denote by τ_{ij}^k the clock offset between nodes Λ_j and Λ_i at the time instant when probe k was transmitted by node Λ_i. We will assume that the frequency difference between any two clocks is small and the clock offset accumulated during a probe exchange is negligible, hence τ_{ij}^k will also denote the offset between the two clocks when probe k is received by node Λ_j.

Let ΔT_{ij}^k be the difference between the receiving time R_{ij}^k and the transmission time T_{ij}^k of probe k, i.e., $\Delta T_{ij}^k = R_{ij}^k - T_{ij}^k$. Note that since the different times are taken according to different clocks which are not necessarily synchronized, ΔT_{ij}^k, is not necessarily the delay but rather the one way link delay experienced by probe k while travelling between nodes Λ_i and Λ_j, plus the difference between the two clocks, $\Delta T_{ij}^k = d_{ij}^k + \tau_{ij}^k$ (see Fig. 2). Note that ΔT_{ij}^k can take negative values.

Finally, let us define the frequency of Λ_0 to be one, and denote the ratio of the frequency of node $\Lambda_i \in \mathcal{N}$ to the frequency of Λ_0 by ω_i. The ratio between the frequencies of any two nodes $\Lambda_i, \Lambda_j \in \mathcal{N}$ will be denoted by $\omega_{ij} \stackrel{\Delta}{=} \frac{\omega_i}{\omega_j}$. The following is an important relation that holds for any two probe packets k and l (see Fig. 3):

$$\omega_{ji} - 1 = \frac{\omega_j - \omega_i}{\omega_i} = \frac{\tau_{ij}^l - \tau_{ij}^k}{T_{ij}^l - T_{ij}^k} \tag{1}$$

or

$$\tau_{ij}^l = \tau_{ij}^k + (\omega_{ji} - 1)(T_{ij}^l - T_{ij}^k)$$

The latter equation describes the evolution of the offset between Λ_i and Λ_j due to frequencies difference between the clocks of the two nodes. Clearly, if $\omega_i = \omega_j$, then the offset is constant.

Since $\omega_{ji} - 1$ denotes the respective slope of the offset evolution, we denote $\omega_{ji} - 1 = \tan(\alpha_{ij})$ (see Fig. 3). It is important to note that $\omega_{ji} - 1 = \tan(\alpha_{ij})$ cannot be measured, since neither τ_{ij}^l nor τ_{ij}^k can be measured. Consequently, in order to estimate ω_{ji}, we will need to estimate $\tan(\alpha_{ij})$.

III. Measurements Filter

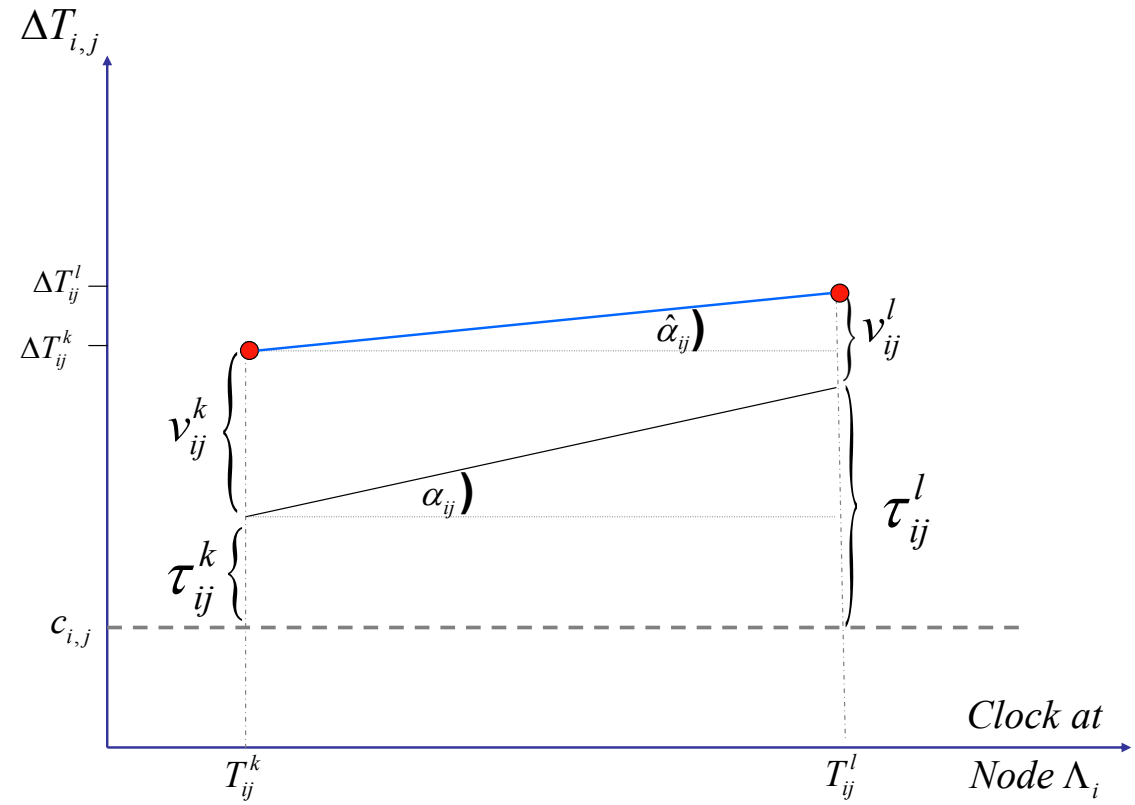
Our global scheme for estimating the frequency difference between each node and the reference node is based on the estimation of pair-wise relative frequency differences between each neighboring pair of clocks (estimation of ω_{ij}). The estimation of these pair-wise relative frequency differences is not based on two measurements as described in Sec. II-B, but rather on a filtered version of many probe measurements (see Fig. 1). In this section we explain the operation of the measurement filter. We focus without loss of generality on the two nodes Λ_i, Λ_j $e_{ij} \in \mathcal{E}$.

The estimation of ω_{ji} is based on estimating the slope $\tan(\alpha_{ij}) = \omega_{ji} - 1$. To that end we examine the following slope denoted by $\tan(\hat{\alpha}_{ij})$ (see Fig. 3) that can be easily computed from the relevant measurements:

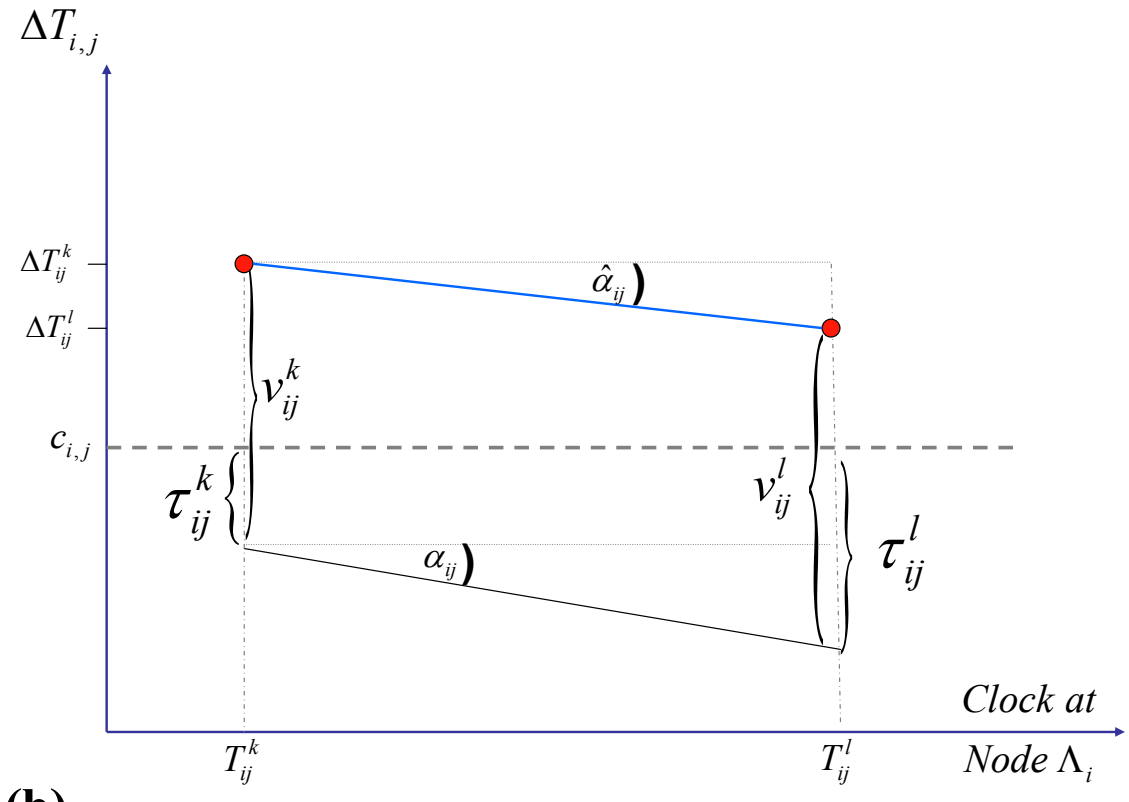
$$\tan(\hat{\alpha}_{ij}) = \frac{\Delta T_{ij}^l - \Delta T_{ij}^k}{T_{ij}^l - T_{ij}^k} \tag{2}$$

Now we will explore the relation between $\tan(\alpha_{ij})$ and $\tan(\hat{\alpha}_{ij})$

$$\begin{aligned}
\tan(\hat{\alpha}_{ij}) &= \frac{\Delta T_{ij}^l - \Delta T_{ij}^k}{T_{ij}^l - T_{ij}^k} = \frac{(R_{ij}^l - T_{ij}^l) - (R_{ij}^k - T_{ij}^k)}{T_{ij}^l - T_{ij}^k} \\
&= \frac{(\overbrace{(T_{ij}^l + c_{ij}^l + v_{ij}^l + \tau_{ij}^l)}^{R_{ij}^l} - T_{ij}^l)}{T_{ij}^l - T_{ij}^k} \\
&\quad - \frac{(\overbrace{(T_{ij}^k + c_{ij}^k + v_{ij}^k + \tau_{ij}^k)}^{R_{ij}^k} - T_{ij}^k)}{T_{ij}^l - T_{ij}^k} \\
&= \frac{\overbrace{(c_{ij}^l - c_{ij}^k)}^{=0} + \overbrace{(v_{ij}^l - v_{ij}^k)}^{\Delta v_{ij}^{l,k}} + \overbrace{(\tau_{ij}^l - \tau_{ij}^k)}^{\Delta \tau_{ij}^{l,k}}}{T_{ij}^l - T_{ij}^k} \\
&= \frac{\Delta v_{ij}^{l,k}}{T_{ij}^l - T_{ij}^k} + \overbrace{\left(\frac{\Delta \tau_{ij}^{l,k}}{T_{ij}^l - T_{ij}^k}\right)}^{\tan(\alpha_{ij})}
\end{aligned} \tag{3}$$



(a)



(b)

Fig. 3. Definitions and notations. (a) for $\omega_i < \omega_j$ which implies $\tau_{ij}^{l} > \tau_{ij}^{k}$, and (b) for $\omega_i > \omega_j$ ($\tau_{ij}^{l} < \tau_{ij}^{k}$). τ_{ij} can take positive or negative values, in (a) $\tau_{ij}^{k} > 0$ and in (b) $\tau_{ij}^{k} < 0$. Note that in (b) the absolute value of τ_{ij}^{k} is depicted

Note that $\frac{\Delta\tau_{ij}^{l,k}}{T_{ij}^{l}-T_{ij}^{k}} = \tan(\alpha_{ij})$ is exactly the relative frequency difference we are looking for, the change in clock offset divided by the time elapsed. Therefore, in order for $\tan(\hat{\alpha}_{ij})$ to be closer to the relative frequency difference the error term $\frac{\Delta v_{ij}^{l,k}}{T_{ij}^{l}-T_{ij}^{k}}$ should be as small as possible. Several procedures exist to choose the two probes k and l out of all probes exchanged between Λ_i and Λ_j, that mostly reduce the error term $\frac{\Delta v_{ij}^{l,k}}{T_{ij}^{l}-T_{ij}^{k}}$. Several procedures are described in [5–7]. The basic idea behind all suggested schemes is to reduce the term $\Delta v_{ij}^{l,k}$ by trying to identify two probes that experience negligible or even no queueing delay ($v_{ij}^{k} \approx 0$ and $v_{ij}^{l} \approx 0$) and use them to determine $\tan(\hat{\alpha}_{ij})$. Alternatively, reducing the error term can be accomplished by enlarging $T_{ij}^{l} - T_{ij}^{k}$, i.e., spreading away the two measurement points. Consequently, if queueing delay is moderate, we can see that if we pick a very long segment, the estimated relative frequency difference will be quite good regardless of the scheme used. We can also conclude from this observation that if we limit the number of probes that we send we should spread them as much as possible.

The relative frequency difference between two neighboring nodes Λ_i and Λ_j is computed based on two different sets of measurements: first for link e_{ij} ($\tan(\hat{\alpha}_{ij})$) and second for e_{ji} ($\tan(\hat{\alpha}_{ji})$). Note that in the NTP standard [16], measurements for both directions are always taken. We expect that the two estimations should concur, as we have that: $\omega_{ji} = \tan(\alpha_{ij}) + 1$, $\omega_{ij} = \tan(\alpha_{ji}) + 1$ and $\omega_{ij} = \frac{1}{\omega_{ji}}$; combining the three we get $\tan(\alpha_{ij}) = \frac{-\tan(\alpha_{ji})}{\tan(\alpha_{ji})+1}$. It is reasonable to expect that $\tan(\hat{\alpha}_{ij}) \approx \frac{-\tan(\hat{\alpha}_{ji})}{\tan(\hat{\alpha}_{ji})+1}$. Since both nodes (in NTP) collect both values, few actions can be taken when the values do not obey the relation $\tan(\hat{\alpha}_{ij}) \approx \frac{-\tan(\hat{\alpha}_{ji})}{\tan(\hat{\alpha}_{ji})+1}$. For instance, we can use both entries for global procedure regardless of that fact. Alternatively, one can filter the links where the estimated frequency difference is greater than some predefined threshold. In case the link is crucial for connectivity we can use both entries, choose one or use some combination. Such a difference can also serve as an indicator to take more measurements until the two slopes get closer.

The inputs for our global procedure are the estimated values of $\tan(\alpha_{ij}) \quad e_{ij} \in \mathcal{E}$ (Fig. 1). Since many schemes can be used we will refer to the estimated $\tan(\alpha_{ij})$ simply by $\tan(\hat{\alpha}_{ij})$ regardless of the measurement filter used.

IV. The Global Optimization Problem

In this section we formulate the frequency synchronization problem as a global optimization problem. The variables are the set of clock adjustments, denoted by $\vec{x} = \{x_1, x_2, \ldots, x_{N-1}\}$, where x_i denotes the relative frequency adjustment that should be performed by node Λ_i, such that if node Λ_i clock frequency is ω_i its frequency after synchronization will be $x_i \cdot \omega_i$. For example $x_i = 1.1$ means that node Λ_i should run its clock 1.1 times faster than it is currently running. Obviously $x_i < 1$ defines a clock that is running fast and should run slower, $x_i > 1$ describes a clock that should run faster, and $x_i = 1$ defines a clock that should be left unchanged (x_0 that relates to the reference frequency is 1 by definition). Since $x_i = 0$ implies that the clock should be stopped, and $x_i < 0$ suggests that the clock is running backward, both unreasonable results, we restrict x_i to be strictly positive, i.e., $x_i > 0$, $\forall \Lambda_i \in \mathcal{N}$. The input for the problem includes all the estimated frequency differences between neighboring nodes, $\tan(\hat{\alpha}_{ij})$ $\forall e_{ij} \in \mathcal{E}$ as explained in section III. To complete the optimization problem description we need to choose a proper objective function.

The ultimate assessment of a synchronization scheme quality is based on the aggregation of all the frequency differences of each clock with respect to the reference frequency. We may try to apply this logic directly by selecting the objective function to be $\min_{\vec{x}} \left[\sum_{\Lambda_i \in \mathcal{N}} |x_i \omega_i - \omega_0|\right]$. However, if we knew how to calculate the frequency difference between each clock and the reference node we

would know how to frequency synchronize the clocks in the first place. Since we cannot precisely measure ω_i using ω_0 as a reference, the objective function should be based on the estimated ω_i's.

Another aspect that should be considered is that communication between any node and the reference node is accomplished via probe forwarding among neighboring nodes. Trying to estimate the frequency differences between a clock and the reference node will rely on the frequency difference between neighboring nodes comprising the path between the two nodes. Since measurement noise (due to queueing delay) is accumulated with each additional link, it seems reasonable to base the objective function on the relative frequency difference or its estimation between neighboring nodes rather than on the frequency difference between each node and the reference node, i.e., the objective function should be a function of ω_{ij} $e_{ij} \in \mathcal{E}$, rather than of ω_i, $\Lambda_i \in \mathcal{N}$.

A natural function of the relative pairwise frequencies that may serves as the objective function is

$$S = Min_{\vec{x}} \left[\sum_{e_{ij} \in \mathcal{E}} \left(\frac{x_j \omega_j}{x_i \omega_i} - 1 \right)^2 \right] \tag{4}$$

The motivation for (4) is clear. One would like that all frequencies, after they are adjusted, would be as close to each other as possible. Since $x_0 = \omega_0 = 1$, it is clear that (4) attempts to bring all the adjusted frequencies close to that of the reference node Λ_0.

Using the fact that $\frac{\omega_j}{\omega_i} = \tan \alpha_{ij} + 1$, (4) can be written as:

$$S = Min_{\vec{x}} \left[\sum_{e_{ij} \in \mathcal{E}} \left(\frac{x_j}{x_i} \left(\tan(\alpha_{ij}) + 1 \right) - 1 \right)^2 \right] \tag{5}$$

However, as explained in Section III, the values $\tan(\alpha_{ij})$ for all $i, j \quad \forall e_{ij} \in \mathcal{E}$ are not available or computable, but can be estimated via the computable quantities $\tan(\hat{\alpha}_{ij})$. Consequently, we can use the respective estimated value in (5).

Now we can complete the formalization of the frequency synchronization problem:

Given $\tan(\hat{\alpha}_{ij}) \quad \forall e_{ij} \in \mathcal{E}$, find the set of frequency adjustments $x_i \quad \forall \Lambda_i \in \mathcal{N}$ that optimizes:

$$S = Min_{\vec{x}} \left[\sum_{e_{ij} \in \mathcal{E}} \left(\frac{x_j}{x_i} \cdot \left(\tan(\hat{\alpha}_{ij}) + 1 \right) - 1 \right)^2 \right] \tag{6}$$

under the constraints $x_i > 0 \quad \forall \Lambda_i \in \mathcal{N}$.

We term the frequency synchronization procedure that resulted from the solution of (6) as TanSq.

Note that the lowest possible value which can be attained in (6) is $S = 0$ and this can happen only if all $\frac{x_j}{x_i} \cdot (\tan \hat{\alpha}_{ij} + 1) = 1$, $\forall e_{ij} \in \mathcal{E}$. Obviously this will be the case when there is no measurement noise (queueing delay) and the modified clocks are exactly synchronized, i.e. $\frac{x_j}{x_i} \cdot (\tan \hat{\alpha}_{ij} + 1) = \frac{x_j}{x_i} \cdot (\tan \alpha_{ij} + 1) = \frac{x_j \omega_j}{x_i \omega_i} = 1$ or $x_j \omega_j = x_i \omega_i$, $\forall e_{ij} \in \mathcal{E}$.

V. Small Frequency Deviation Approach (SFD)

In the previous section we introduced the frequency synchronization problem as a network-wide optimization problem. The objective function suggested is aimed to minimize the relative frequency differences between each node and its neighbors. Numerical results are provided in Section VI. In this section we present a simpler approach in terms of computational complexity, which under practical circumstances, when the frequency deviations of the nodes are small, gives results nearly as good as the one described in (6).

Denoting the relative frequency difference between clocks of Λ_0's and Λ_i's with respect to the reference frequency by y_i, i.e. $y_i = \frac{\omega_0 - \omega_i}{\omega_0} = 1 - \omega_i$. Using the Taylor series expansion of $\frac{\omega_j - \omega_i}{\omega_i}$ about $(\omega_i, \omega_j) = (1, 1)$ we get:

$$\begin{aligned} \frac{\omega_j - \omega_i}{\omega_i} &= \frac{(1 - y_j) - (1 - y_i)}{(1 - y_i)} = \frac{-(y_j - y_i)}{(1 - y_i)} \\ &= y_i - y_j + O(y_i^2, y_i y_j) \end{aligned} \tag{7}$$

which is the frequency difference between Λ_j and Λ_i clocks according to the reference frequency, plus terms of the frequency difference of higher order. This means that if the starting frequencies of the nodes are relatively close to $\omega_0 = 1$, then $\frac{\omega_j - \omega_i}{\omega_i}$ is a good estimation for the frequency difference between the two according to Λ_0 time units, i.e., $\frac{\omega_j - \omega_i}{\omega_i} \approx \frac{\omega_j - \omega_i}{\omega_0}$. For example, assume two nodes where the first clock accumulates ten more minutes per day than Λ_0 and the other clock accumulates ten minutes less per day than Λ_0, we get: $\frac{\omega_j - \omega_i}{\omega_i} = \frac{20}{1450} = 0.0138$; taking the approximated value we get: $\omega_j - \omega_i = \frac{20}{1440} = 0.0139$.

Using the last approximation in Eq. (3) we get:

$$\begin{aligned} \tan(\hat{\alpha}_{ij}) &= \tan(\alpha_{ij}) + \frac{\Delta v_{ij}^{l,k}}{T_{ij}^l - T_{ij}^k} \\ &= y_i - y_j + \frac{\Delta v_{ij}^{l,k}}{T_{ij}^l - T_{ij}^k} + O(y_i^2, y_i y_j) \\ &= (\omega_j - \omega_i) + \frac{\Delta v_{ij}^{l,k}}{T_{ij}^l - T_{ij}^k} + O(y_i^2, y_i y_j) \end{aligned} \tag{8}$$

To conclude, we can say that when the starting frequencies are relatively close to one, $\tan(\hat{\alpha}_{ij})$ is a good approximation for $\omega_j - \omega_i$. It is important to emphasize that the measurement noise is as before: $\frac{\Delta v_{ij}^{l,k}}{T_{ij}^l - T_{ij}^k}$.

Note that $\vec{y} = \{y_1, y_2, \ldots, y_{N-1}\}$ is the set of additive frequency adjustments that should be performed by the respective nodes, i.e., the clock frequency of node Λ_i after synchronization would be $\omega_i + y_i$.

A natural objective function in this case is:

$$S = Min_{\vec{y}} \left[\sum_{e_{ij} \in \mathcal{E}} (\omega_j + y_j - \omega_i - y_i)^2 \right] \tag{9}$$

The motivation for (9) is similar to that of (4), except that here we take the frequency difference between all pair of nodes according to the reference node time units.

Since $\tan(\hat{\alpha}_{ij})$ is a good estimation of $\omega_j - \omega_i$ (see (8)), we conclude that the Small Frequency Deviation (SFD) refinement can be formalized as:

$$S = Min_{\vec{y}} \left[\sum_{e_{ij} \in \mathcal{E}} (\tan(\hat{\alpha}_{ij}) + y_j - y_i)^2 \right] \tag{10}$$

where $\tan(\hat{\alpha}_{ij}) \quad \forall e_{ij} \in \mathcal{E}$, are given from measurements, and $y_i \quad \forall \Lambda_i \in \mathcal{N}$ are the variables to be optimized under the constraints that $y_i > 0 \; \forall \Lambda_i \in \mathcal{N}$.

There are a few advantages of using the SFD objective function. The first major advantage is that by using this method it is easy to find the point $\vec{y}$ that yields the minimum analytically.

Let us partially differentiate (10) with respect to each variable, y_i $\forall \Lambda_i \in \{\mathcal{N} \setminus \Lambda_0\}$ ($y_0 = 0$ by definition) and equate it to zero.

$$\begin{aligned}
\frac{\partial S}{\partial y_i} &= \frac{\partial}{\partial y_i}\left(\sum_{e_{ij}\in\mathcal{E}} (\tan(\hat{\alpha}_{ij}) + y_i - y_j)^2\right) \\
&= 2 \cdot \left(\sum_{\{l|e_{il}\in\mathcal{E}\}} (\tan(\hat{\alpha}_{i,l}) + y_i - y_l)\right. \\
&\qquad \left. - \sum_{\{l|e_{li}\in\mathcal{E}\}} (\tan(\hat{\alpha}_{l,i}) + y_l - y_i)\right) \\
&= 2 \cdot \sum_{\{l|e_{il}\in\mathcal{E}\}} ((\tan(\hat{\alpha}_{i,l}) - \tan(\hat{\alpha}_{l,i})) + 2y_i - 2y_l) \\
&= 0
\end{aligned} \tag{11}$$

The equation set described in (11) can be written as:

$$|G_i| \cdot y_i - \sum_{\{l|e_{il}\in\mathcal{E}\}} y_l = \sum_{\{l|e_{il}\in\mathcal{E}\}} \frac{(\tan(\hat{\alpha}_{i,l}) - \tan(\hat{\alpha}_{l,i}))}{2} \qquad \forall \Lambda_i \in \{\mathcal{N} \setminus \Lambda_0\} \tag{12}$$

It is interesting to note that based on (11), in the optimal solution the sum of all the modified $\tan(\hat{\alpha}_{i,l})$ related to each node is equal to zero, i.e.,
$\sum_{\{l|e_{il}\in\mathcal{E}\}} ((\tan(\hat{\alpha}_{i,l}) - \tan(\hat{\alpha}_{l,i})) + 2y_i - 2y_l) = 0$.

Another important advantage of the SFD objective function is that it can be easily implemented in a distributed fashion. There is more than one method that can be used to solve the equation set suggested in (12) over a distributed environment [17], [18]. In the sequel we present a distributed protocol that converges to the optimal frequency values attained in (12).

The basic structure of the distributed SFD approach is that each node Λ_i, besides node Λ_0, maintains a record in which it holds the entries $\hat{\alpha}_{ij}$ and $\hat{\alpha}_{ji}$ for each neighbor $\Lambda_j \in G_i$. In order to maintain the record, each node periodically transmits a probe over each of its outgoing links, performs the measurement filter on the measurements and changes its record accordingly.

The suggested distributed optimization is iterative. In each iteration, a subset of nodes, which can include any number of nodes between one node to all nodes beside Λ_0, performs a Frequency Refinement Procedure. According to this procedure, each node adjusts its clock by $y_i = \frac{1}{|G_i|}\sum_{j\in G_i} \frac{(\tan(\hat{\alpha}_{ij}) - \tan(\hat{\alpha}_{ji}))}{2}$. After each clock adjustment, node Λ_i modifies its records, $\tan^{new}(\hat{\alpha}_{ij}) = \tan^{old}(\hat{\alpha}_{ij}) + y_i$ and $\tan^{new}(\hat{\alpha}_{ji}) = \tan^{old}(\hat{\alpha}_{ji}) - y_i$. In addition, it transmits its clock change to all its neighbors. When node Λ_j receives a notification regarding a clock change performed by one of its neighbors, it modifies the record entries related to this node, $\tan^{new}(\hat{\alpha}_{ji}) = \tan^{old}(\hat{\alpha}_{ji}) - y_i$ and $\tan^{new}(\hat{\alpha}_{ij}) = \tan^{old}(\hat{\alpha}_{ij}) + y_i$. Note that the total record changes performed after each iteration due to the clocks adjustments in both node Λ_i and Λ_j clocks are $\tan^{new}(\hat{\alpha}_{ij}) = \tan^{old}(\hat{\alpha}_{ij}) + y_i - y_j$ and $\tan^{new}(\hat{\alpha}_{ji}) = \tan^{old}(\hat{\alpha}_{ji}) - y_i + y_j$.

Theorem 1: When the SFD frequency adjustment operation is applied by all nodes in all iterations, the set of frequencies converges to the set of frequencies which minimizes the objective

function (10), and each clock in the network will converge eventually to the clock that would have been obtained by solving the equation set described in (12).
The proof appears in the Appendix.

VI. Numerical Analysis

In sections IV and V we described two global procedures for frequency synchronizing clocks in a network. In this section we will evaluate and compare the two different methods by simulation.

In order to stay as close as possible to real network characteristics we constructed the network topology setup and chose the relevant parameters based on the existing extensive literature of modelling network topologies and delays.

A. Underlying Topology and Delay

In order to evaluate the frequency synchronization schemes, we first need to construct the network topology setup. As explained in Section II-A, we focus on an overlay network that consists of the components that participate in the frequency synchronization procedure hence we need to model and characterize the overlay network of synchronizing entities. Our network topology is based on the random model [19]. Other parameters such as delay processes and parameters, clock offsets etc. were chosen based on the literature [20], [21], [22], [14], [23].

The random network construction is based on the Breadth First Search (BFS) principle. We start with a single Reference Time Node, restrict the hop distance of each node to the Reference Time Node to be at most a certain number of hops. The depth (maximum hop distance from Λ_0) is selected to be 5 according to [21] which studies the deployment of NTP structure hierarchy. The links between the nodes (within the same hierarchy or between adjacent layers) are randomly selected. The delays are assumed to be distributed according to shifted Erlang distribution which best represents Internet one way path delay [22], [14], [23]. The shifted Erlang parameters were selected according to the IEEE 802.20 Working Group [20], and found to be compliant with [21]. The shift associated with the propagation delay of each link is chosen based on uniform distribution ($\sim U[0, 10]$), the number of exponentials (α) and the mean time between events (θ), are randomly selected between 1 and 5 and between 0.1 and 3, respectively. The parameters are sampled once for each directed link. Clock offsets with respect to the reference frequency were selected based on [21] to randomly vary with a uniform distribution between -10 and 10 ($\sim U[-10, 10]$). Clock frequencies are uniformly selected in a symmetric region around ω_0.

Thirty probes were sent over each link and the measurement filter was applied on these probes. The time difference between two consecutive probes was randomly selected with uniform distribution, where the uniform interval was changed between the different runs. The measurement filter used for both the SFD as well as the TanSq is based on the second scheme suggested in [7]. According to this scheme the slope of the line that minimizes the area between itself and the serrated line connecting all measurement points (the 2-tuples $(T_{ij}^k, \Delta T_{ij}^k)$), is the estimated frequency difference between the two nodes Λ_i and Λ_j according to node Λ_i, i.e. ω_{ji}. It is important to note that as explained in Section III both procedures TanSq as well as SFD work well using all measurement filters suggested [5]. The first scheme suggested in [7] which is based on [5] was also tested and yields the same result patterns. Also note that the measurement filter suggested in Section III which takes two measurements taken at distant time points can also be used.

B. Results

In order to evaluate the two global schemes suggested in Sections IV and V we suggest a third global frequency synchronization scheme which will be evaluated with respect to the other two schemes. The third scheme, termed the Hierarchical Frequency Synchronization Protocol (HFSP),

takes the hierarchical approach in the spirit of the NTP protocol. According to HFSP each node arbitrarily selects a single neighbor which is one hop closer to the Reference Time Node than itself and frequency synchronize with respect to it. The relative frequency difference between the two nodes is computed based on the second scheme suggested in [7].

In order to thoroughly examine the results of the three schemes, the two suggested global schemes suggested in Section IV and Section V which exploit convex optimization techniques, and the HFSP which is a hierarchical approach, are simulated over a wide variety of network topologies and scenarios.

We started by examining the three schemes over two random networks comprising 252 and 998 nodes. The initial starting frequency of each node varied uniformly between 0.7 to 1.3 times the reference frequency, i.e. $\omega_i \sim U[0.7, 1.3]$.

Figure 4 demonstrates the frequency dispersion around the reference frequency in a 252 node network and in a 998 node network, respectively. In these graphs, the x axis is the node ID. The y axis is the frequency with respect to the reference frequency after performing each scheme.

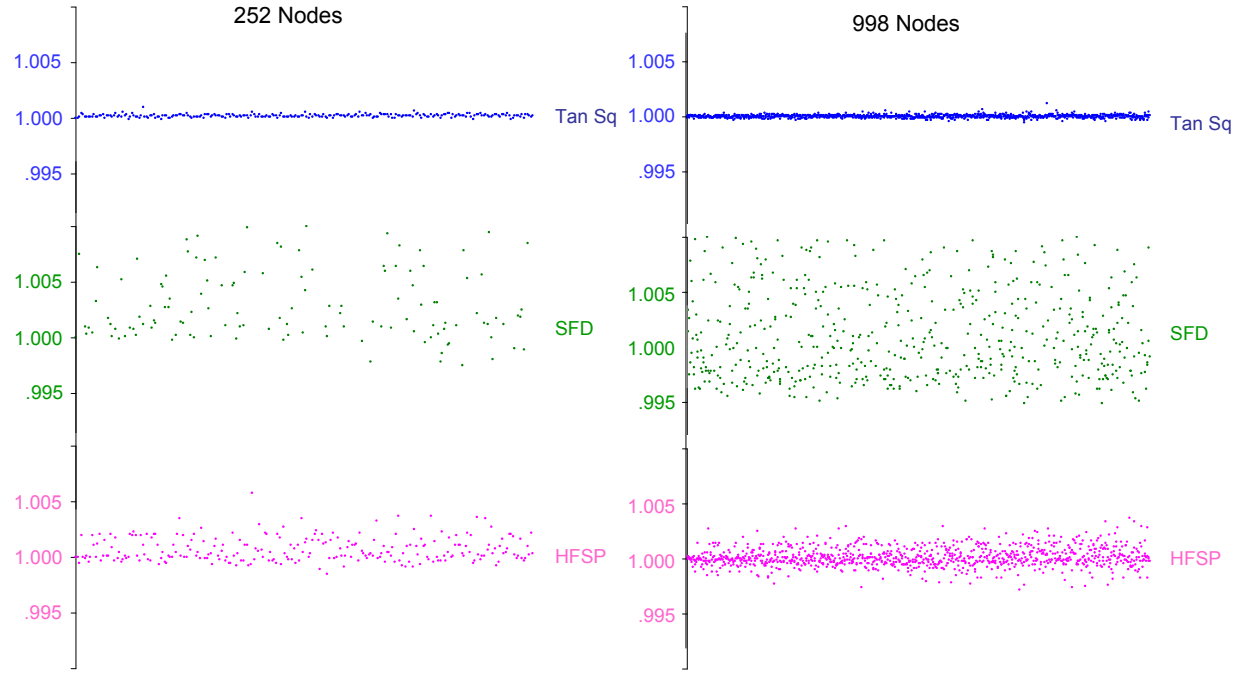


Fig. 4. Frequency dispersion around the reference frequency on 252 and 998 node networks, for initial starting frequency of each node that varied uniformly between 0.7 to 1.3 the reference frequency

As can be seen in the figures all three schemes significantly improve the initial frequency (note that some of the nodes had initial frequencies around 0.7 and 1.3). However it can also be seen in the figure that as expected since the initial frequency range is quite large ($\omega_i \sim U[0.7, 1.3]$), both the TanSq and the HFSP perform much better than the SFD approach. In Figure 5 we present the same results in a different way. In this figure we present the distribution of the absolute value of the frequency difference with respect to the reference frequency, i.e., the y axis depicts the value $Prob[|\omega_0 - \omega_i| \leq x]$ after the frequency synchronization. We also added to the graphs the distribution of the initial frequencies which is denoted by Original.

It can be seen that the TanSq works very well even when the initial frequencies are quite different from the reference frequency. For example using TanSq 99.2% of the clocks are less than 0.0006 apart from the reference frequency, and all the clocks are less than 0.0011 times the reference

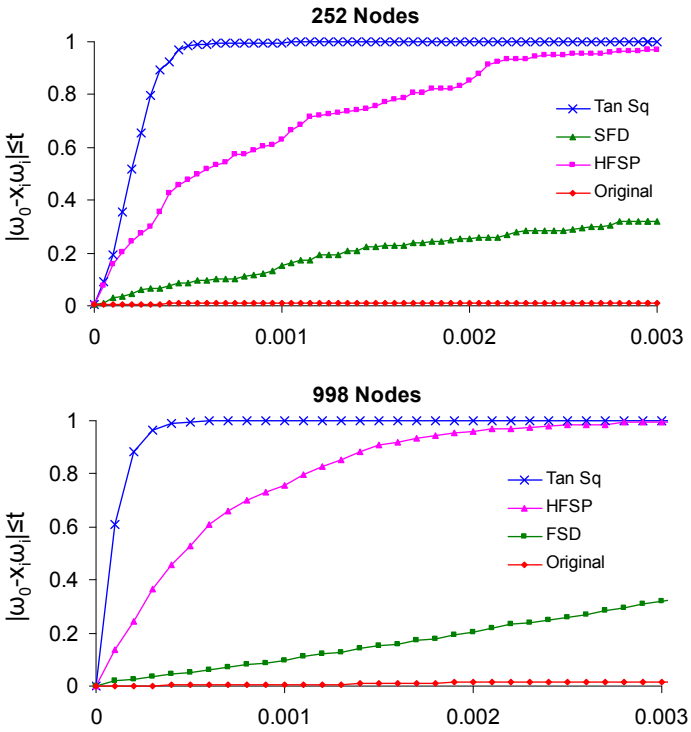


Fig. 5. The frequency offset distribution on 252 and 998 node networks

frequency in the 252 node network, while using the other two schemes we get only 51.6% and 9.5% of the nodes less than 0.0006 apart from the reference frequency in the HFSP and SFD respectively, and only 68.7% and 17.5% less than 0.0011 apart from the reference frequency in the HFSP and SFD respectively. Note that only 0.8% of the nodes start with a frequency which is 0.0011 apart from the reference frequency. In the 998 node network the results are quite similar, all nodes are less than 0.0012 from the reference frequency after performing the TanSq while only 85% and 13% are less than 0.0012 from the reference frequency using the HFSP and SFD respectively. Again only 0.8% of the nodes start with a frequency which is 0.0012 apart from the reference frequency.

In the next two examples we ran the three schemes over 224 node network and over 505 node network where the initial frequencies vary in the range $U[0.999, 1.001]$.

Figures 6 and 7 demonstrate the frequency dispersion around the reference frequency on the 224 and on 505 node networks and the distribution of the absolute value of the frequency difference with respect to the reference frequency respectively.

Both graphs 6 and 7 clearly demonstrate that when the initial frequency of the nodes is close to the reference frequency, both schemes, SFD and TanSq, perform very well. For example in the 224 node network in both schemes all frequencies are less than 0.0006 apart from the reference frequency, while only 82% of the nodes are 0.0006 apart from reference frequency after performing the HFSP. In the 505 node network the results are about the same. 100%, 99.6% and 48.5% of the nodes have their frequency less than 0.00065 apart from the reference frequency after performing the TanSq, SFD and HFSP schemes respectively.

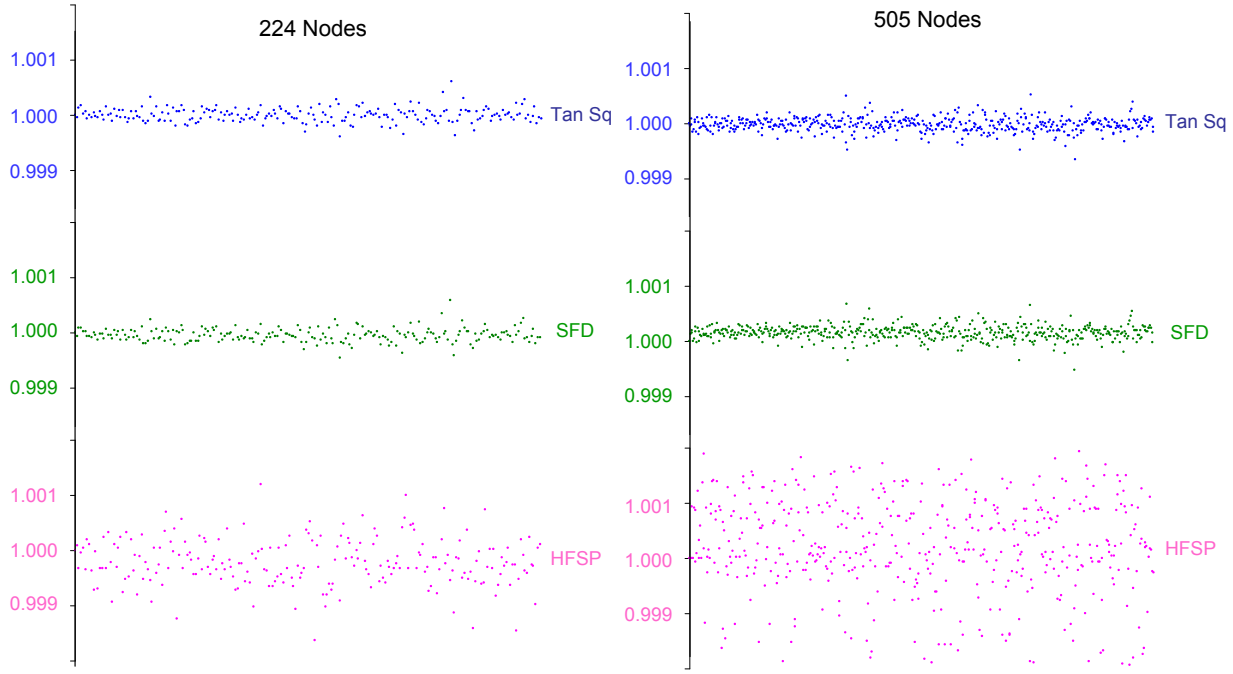


Fig. 6. Frequency dispersion around the reference frequency on 224 and 505 node networks; the initial starting frequency of each node varied uniformly between 0.999 to 1.001 times the reference frequency

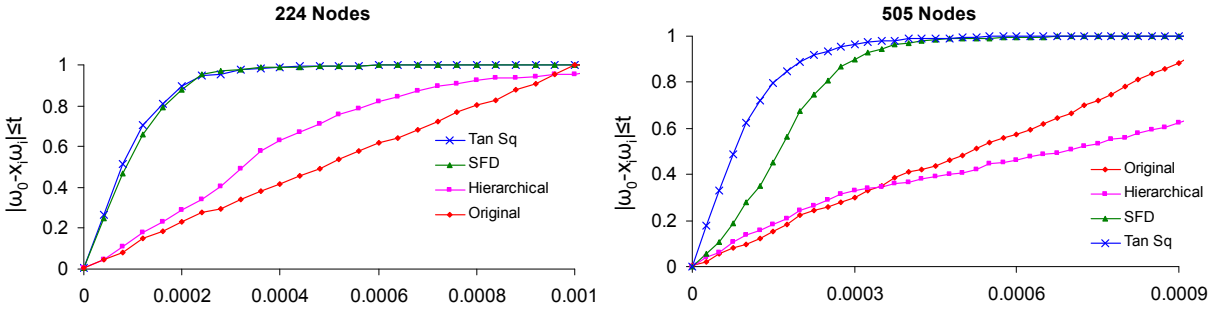


Fig. 7. The frequency offset distribution on 224 and 505 node networks

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Appendix

Proof of Theorem 1:

The proof to theorem 1 comprises two propositions. The first proposition claims that the series of values obtained by the objective function after each iteration converge. The second proposition states that the value the objective function converges into is the optimal value, and the frequencies resulted by running the distributed algorithm is the same as the one obtained by solving the equation set (12).

We will use the superscript $[h]$ for all values that relate to the h-th iteration. For instance, $y_i^{[h]}$ denotes the frequency adjustment performed by node Λ_i in the h-th iteration, $\tan^{[h]}(\hat{\alpha}_{ij})$ denotes the value of $\tan(\hat{\alpha}_{ij})$ after the h-th iteration, etc.

Proposition 1: If a set of arbitrary nodes, denoted by Ψ, adjust their frequency by $y_i^{[h]} = \frac{1}{|G_i|}\sum_{j\in G_i}\frac{\left(\tan^{[h-1]}(\hat{\alpha}_{ij})-\tan^{[h-1]}(\hat{\alpha}_{ji})\right)}{2}$, the new sum $\sum_{\forall e_{ij}\in\mathcal{E}}\left(\tan^{[h]}(\hat{\alpha}_{ij})\right)^2$ is not bigger than the sum prior to the adjustment.

Proof of Proposition 1: Since nodes belong to set Ψ modify their frequency by $\frac{1}{|G_i|}\sum_{j\in G_i}\frac{(\tan(\hat{\alpha}_{ij})-\tan(\hat{\alpha}_{ji}))}{2}$

and nodes which are not in the set do not adjust their frequency at all, let us denote:

$$y_i^{[h]} = \begin{cases} \frac{1}{|G_i|}\sum_{j\in G_i} \frac{\left(\tan^{[h-1]}(\hat{\alpha}_{ij}) - \tan^{[h-1]}(\hat{\alpha}_{ji})\right)}{2} & \text{if } \Lambda_i \in \Psi \\ 0 & \text{otherwise} \end{cases} \tag{13}$$

Let us denote the objective function by S:

$$\begin{aligned} S^{[h]} &= \sum_{e_{ij}\in\mathcal{E}} \left(\tan^{[h-1]}(\hat{\alpha}_{ij}) - y_i^{[h]} + y_j^{[h]}\right)^2 \\ &= \sum_{e_{ij}\in\mathcal{E}} \Big((\tan^{[h-1]}(\hat{\alpha}_{ij}))^2 - 2(\tan^{[h-1]}(\hat{\alpha}_{ij}))y_i^{[h]} + 2(\tan^{[h-1]}(\hat{\alpha}_{ij}))y_j^{[h]} \\ &\qquad +(y_i^{[h]})^2 + (y_j^{[h]})^2 - 2y_i^{[h]}y_j^{[h]}\Big) \\ &= \sum_{e_{ij}\in\mathcal{E}} \left(\tan^{[h-1]}(\hat{\alpha}_{ij})\right)^2 - \sum_{e_{ij}\in\mathcal{E}} \left(y_i^{[h]} + y_j^{[h]}\right)^2 \\ &+ \sum_{e_{ij}\in\mathcal{E}} \left(2(y_i^{[h]})^2 + 2(y_j^{[h]})^2 - 2y_i^{[h]}\tan^{[h-1]}(\hat{\alpha}_{ij}) + 2y_j^{[h]}\tan^{[h-1]}(\hat{\alpha}_{ij})\right) \end{aligned} \tag{14}$$

Let us concentrate on the third part and instead of summing on each link twice for each direction separately, let us sum once for both directions:

$$\begin{aligned} &\sum_{e_{ij}\in\mathcal{E}} \left(2(y_i^{[h]})^2 + 2(y_j^{[h]})^2 + 2(\tan^{[h-1]}(\hat{\alpha}_{ij}))y_i^{[h]} - 2(\tan^{[h-1]}(\hat{\alpha}_{ij}))y_j^{[h]}\right) \\ &= 2\sum_{\substack{e_{ij}\in\mathcal{E}\\ i<j}} \Big\{\left((y_i^{[h]})^2 + (y_j^{[h]})^2 + (\tan^{[h-1]}(\hat{\alpha}_{ij}))y_i^{[h]} - (\tan^{[h-1]}(\hat{\alpha}_{ij}))y_j^{[h]}\right) \\ &\qquad + \left((y_j^{[h]})^2 + (y_i^{[h]})^2 + (\tan^{[h-1]}(\hat{\alpha}_{ji}))y_j^{[h]} - (\tan^{[h-1]}(\hat{\alpha}_{ji}))y_i^{[h]}\right)\Big\} \\ &= 2\sum_{\substack{e_{ij}\in\mathcal{E}\\ i<j}} \Big\{\left(2(y_i^{[h]})^2 - (y_i^{[h]})(\tan^{[h-1]}(\hat{\alpha}_{ij}) - \tan^{[h-1]}(\hat{\alpha}_{ji}))\right) \\ &\qquad + \left(2(y_j^{[h]})^2 - (y_j^{[h]})(\tan^{[h-1]}(\hat{\alpha}_{ji}) - \tan^{[h-1]}(\hat{\alpha}_{ij}))\right)\Big\} \\ &= 2\sum_{\Lambda_i\in\mathcal{N}}\sum_{\Lambda_j\in G_i} \left\{2(y_i^{[h]})^2 - (y_i^{[h]})\left(\tan^{[h-1]}(\hat{\alpha}_{ij}) - \tan^{[h-1]}(\hat{\alpha}_{ji})\right)\right\} \\ &= 2\cdot\sum_{\Lambda_i\in\mathcal{N}} (y_i^{[h]}) \left\{2|G_i|\frac{1}{|G_i|}\sum_{j\in G_i}\frac{(\tan(\hat{\alpha}_{ij}) - \tan(\hat{\alpha}_{ji}))}{2} - \sum_{j\in G_i}(\tan(\hat{\alpha}_{ij}) - \tan(\hat{\alpha}_{ji}))\right\} = 0 \end{aligned} \tag{15}$$

Note that (16) is zero both for $y_i^{[h]} = \frac{1}{|G_i|}\sum_{j\in G_i}\frac{\left(\tan^{[h-1]}(\hat{\alpha}_{ij}) - \tan^{[h-1]}(\hat{\alpha}_{ji})\right)}{2}$ as well as for $y_i^{[h]} = 0$. Putting it back to (14) we have:

$$\begin{aligned} S^{[h]} &= \sum_{e_{ij}\in\mathcal{E}} \left(\tan^{[h-1]}(\hat{\alpha}_{ij})\right)^2 - \sum_{e_{ij}\in\mathcal{E}} \left(y_i^{[h]} + y_j^{[h]}\right)^2 \\ &= S^{[h-1]} - \sum_{e_{ij}\in\mathcal{E}} \left(y_i^{[h]} + y_j^{[h]}\right)^2 \leq S^{[h-1]} \end{aligned} \tag{16}$$

This means that unless all $y_i^{[h]}$'s are zero the sum after the next iteration is smaller than prior to the iteration.

Proposition 1 suggests that the series $S^{[h]} = \sum_{\forall e_{ij} \in \mathcal{E}} \left(\tan^{[h]}(\hat{\alpha}_{ij})\right)^2$ obtained in each iteration, is a non increasing series. Since the series is bounded from below by zero, it must converge. In proposition 2 we will show that it converges to the optimal value which results by solving (12).

Proposition 2: When the SFD frequency adjustment operation is applied by all nodes in all iterations, the set of frequencies converges to the set of clocks which minimizes the objective function (10), i.e., the set of frequencies that would have been obtained by solving the equation set (12).

Proof of Proposition 2:

Let us assume that the series $S^{[h]}$ converges to some value Θ, and prove that:

$$\Theta = \min_{\vec{y} \in \mathbb{R}^{N-1}} \left\{ \sum_{\forall e_{ij} \in \mathcal{E}} \left(\tan(\hat{\alpha}_{ij}) + y_j - y_i\right)^2 \right\} \tag{17}$$

Based on Proposition 1 and according to (16):

$$\begin{aligned} S^{[h]} &= S^{[h-1]} - \sum_{e_{ij} \in \mathcal{E}} \left(y_i^{[h]} + y_j^{[h]}\right)^2 \overset{h\to\infty}{\longrightarrow} \Theta \\ &\Rightarrow \quad \sum_{e_{ij} \in \mathcal{E}} \left(y_i^{[h]} + y_j^{[h]}\right)^2 \overset{h\to\infty}{\longrightarrow} 0 \\ &\Rightarrow \quad y_i^{[h]} + y_j^{[h]} \overset{h\to\infty}{\longrightarrow} 0 \qquad \forall e_{ij} \in \mathcal{E} \end{aligned}$$

Since $y_0 = 0$ we have:

$$\begin{aligned} y_0 = 0 \quad &\Rightarrow \quad y_i^{[h]} \overset{h\to\infty}{\longrightarrow} 0 \qquad \forall \Lambda_i \in G_0 \\ &\Rightarrow \quad y_j^{[h]} \overset{h\to\infty}{\longrightarrow} 0 \qquad \forall \Lambda_j \in G_{\Lambda_i \in G_0} \\ &\Rightarrow \quad \cdots \quad \Rightarrow \quad y_i^{[h]} \overset{h\to\infty}{\longrightarrow} 0 \qquad \forall \Lambda_i \in N \\ &\Rightarrow \quad y_i^{[h]} = \frac{1}{|G_i|} \sum_{j \in G_i} \frac{\left(\tan^{[h-1]}(\hat{\alpha}_{ij}) - \tan^{[h-1]}(\hat{\alpha}_{ji})\right)}{2} = 0 \qquad \forall \Lambda_i \in \mathcal{N} \setminus \Lambda_0 \\ &\Rightarrow \quad \sum_{j \in G_i} \frac{\left(\tan^{[h-1]}(\hat{\alpha}_{ij}) - \tan^{[h-1]}(\hat{\alpha}_{ji})\right)}{2} = 0 \qquad \forall \Lambda_i \in \mathcal{N} \setminus \Lambda_0 \end{aligned}$$

The consequence of all nodes adjusting their clocks according to the optimal solution which is unique, is that for each node $\sum_{j \in G_i} \frac{\left(\tan^{[h-1]}(\hat{\alpha}_{ij}) - \tan^{[h-1]}(\hat{\alpha}_{ji})\right)}{2} = 0$, consequently $\vec{y}$ must be the optimal solution. *Q.E.D.*